

29/10

		2		
		b_1	b_2	
1	a_1	pq	$p(1-q)$	p
	a_2	$(1-p)q$	$(1-p)(1-q)$	$1-p$
		q	$1-q$	

$$\begin{aligned}
 u_i(p; 1-p; q; 1-q) &= p[q u_i(a_1, b_1) + (1-q) u_i(a_1, b_2)] \\
 &\quad + (1-p)[q u_i(a_2, b_1) + (1-q) u_i(a_2, b_2)] \\
 &= p E_1(a_1, \alpha_2) + (1-p) E_1(a_2, \alpha_2) \\
 \alpha_2 &= (q, 1-q)
 \end{aligned}$$

now the absolute value of payoff isn't important.

ie not ordinal preference.

Eg - stag hunt

		2		
		St	H	
1	St	3, 2	0, 1	
	H	1, 0	1, 1	

(A)

		2		
		St	H	
1	St	4, 4	0, 1	
	H	1, 0	1, 1	

(B)

earlier we said both the games are same by the logic of ordinal preference

l_1	l_2	
$\frac{1}{2}$	0	(st, st)
$\frac{1}{2}$	0	(st, n)
0	1	(n, st)
0	0	(n, n)

for Player 1 $l_1 \sim l_2 \quad q \cap A$

$$\text{for } l_1 \Rightarrow \frac{1}{2} \times (2) + \frac{1}{2} \times (0) + 0(1) + 0(0) = 1$$

$$l_2 \Rightarrow 0(1) + 0(0) + 1(1) + 0(0) = 1$$

$l_1 > l_2 \quad q \cap B$

$$\frac{1}{2}(4) + \frac{1}{2}(0) + 0(1) + 0(0) = 2$$

$$0(1) + 0(0) + 1(1) + 0(0) = 1$$

Mixed Strategy nash equilibrium

Mixed Strategy of a player i is a probability distribution over her action set.

eg - (0.4, 0.6)

σ_i = Mixed strategy of i

probability
distribution

$$\sigma_i(a_k) = p, \quad 0 \leq p \leq 1, \quad a_k \in A_i$$

$$\sum_{a_k \in A_i} \sigma_i(a_k) = 1$$

Mixed strategy profile

$(\alpha_1, \alpha_2, \dots, \alpha_n) = \alpha$ combination of different probability distribution

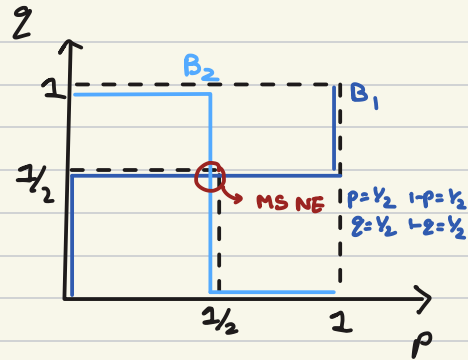
Mixed strategy equilibrium (MSE)

α^* is a mixed strategy NE if for all i $U_i(\alpha^*) \geq U_i(\alpha_i, \alpha_{-i}^*)$, for all α_i of player i

Game of matching pennies

		2			
		H	T		
1	H	1,-1	-1,1	$\left. \begin{matrix} p \\ 1-p \end{matrix} \right\}$	α_1
	T	-1,1	1,-1		
		$\underbrace{\quad 2 \quad \quad 1-2 \quad}_{\alpha_2}$			

$$\begin{aligned}
 U_1(\alpha) &= p[E_1(H, \alpha_2)] + (1-p)[E_1(T, \alpha_2)] \\
 &= p[q(1) + (1-q)(-1)] + (1-p)[q(-1) + (1-q)(1)] \\
 &= p[2q-1] + (1-p)[1-2q]
 \end{aligned}$$



Optimal p^* $\Rightarrow p(2q-1) + (1-p)(1-2q) = \text{OPTIMAL}$
 $2q-1 > 1-2q$ then $p=1$
 $4q > 2$
 $q > 1/2$

If $q < 1/2$ then $p=0$

If $q = 1/2$ then $0 \leq p \leq 1$

for 1

If $q = 1/2$
 $E_1(u_1, \alpha_2) = E_2(u_1, \alpha_2)$
hence

Optimal p^* is
 $0 \leq p \leq 1$

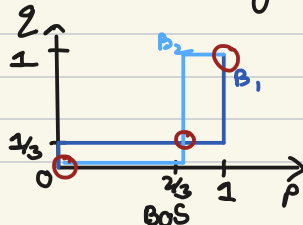
for $p < 1/2$ $q=1$

$p > 1/2$ $q=0$

$p = 1/2$ $0 \leq q \leq 1$

for 2

apply the same logic for BOS?



same NE as earlier
+
some new MSNE

$(0,0)$ $(1,1)$ $(1/3, 1/3)$